

Live Causal Navigator

Pneumococcal vaccination and pneumonia

Module F · Standalone workshop

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Andy Wilson

MODULE D → MODULE F

Same causal logic.
Now every decision is
visible, revisable, and
exportable.

Workshop stance

The Navigator is documentation infrastructure—not an oracle.

CAUSAL INFERENCE FOR REAL-WORLD DATA

A Causal Navigator

Causal inference requires three ingredients.
Our tools help you formalize each one rigorously.

We endorse the **question-first** approach

We align with the philosophy of **Targeted Learning** as articulated by Mark van der Laan: start with what you're trying to learn from the data, not with convenient statistical activities

→ **Question-First**

- Q Start with your causal question
- D Assess your available data
- M Formalize your assumptions

Know if your question is answerable before you analyze

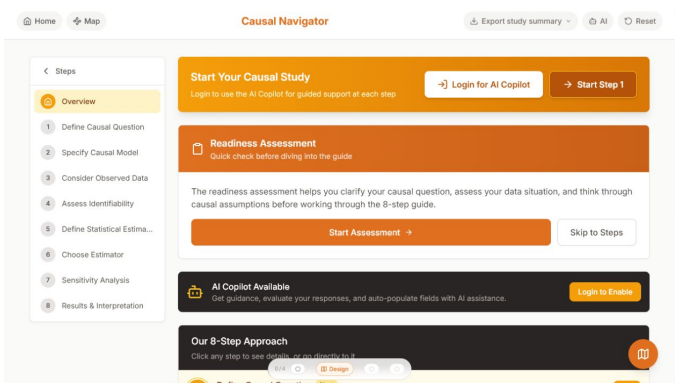
× NOT activity-first

1. Start with data
2. Run statistical tests
3. *Hope results are meaningful*

Risk of answering the wrong question or finding spurious associations

When the three ingredients are specified, your answer will have a **causal interpretation**

Q D M Causal



- **Start at Step 1.** Use the manual route.
- **Work in order.** Pause at design decisions.
- **Export at Step 8.** Keep the Markdown audit trail.

AI support is optional and is not used in this workshop.

navigator.tao-rwd.com

Observed 12-month pneumonia risk

7.19%

vaccinated

6.43%

unvaccinated

Crude RD

+0.76 percentage points

Individual prediction → pair check

Before seeing the prepared answer, name the population, time zero, outcome window, and contrast needed to interpret this difference.

Higher-risk adults are more likely to be vaccinated. The association mixes baseline risk with the treatment effect.

Population	Synthetic adult cohort, $N = 5,000$; age centered at 65 (not restricted to age 65+)
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Treatment	Vaccination at baseline ($A = 1$)
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Comparator	No vaccination at baseline ($A = 0$)
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Outcome	Pneumonia hospitalization within 12 months
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Contrast	Average 12-month risk difference
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Time-zero rule

Eligibility, treatment assignment, and follow-up all begin at the same time zero.

1 Define the causal question

- intervention, comparator, and time zero
- population and eligibility
- outcome horizon
- causal contrast

Checkpoint

Which single phrase would most change the estimand?

The screenshot shows the 'Causal Navigator' web application. On the left is a 'Steps' sidebar with a list of steps: Overview, Define Causal Question (highlighted with a yellow background and a red '1'), Specify Causal Model, Consider Observed Data, Assess Identifiability, Define Statistical Estima..., Choose Estimator, Sensitivity Analysis, and Results & Interpretation. The main content area is titled 'Causal Navigator' and has buttons for 'Export study summary', 'AI', and 'Reset'. It contains three text input fields: 'Outcome' with the text 'Hospitalization with pneumonia within 12 months of time zero (binary Y).', 'Follow-up Period' with the text 'Start at the aligned index date (vaccination or eligible non-vaccination date) and end at 365 days. Censor at death, disenrollment, loss to follow-up, or database end. The teaching simulation has complete 1-year outcomes.', and 'Summary Measure (Causal Contrast)' with the text 'Primary: 1-year risk difference, $E[Y^*1] - E[Y^*0]$. Secondary: 1-year risk ratio, $E[Y^*1] / E[Y^*0]$.' At the bottom, there is a navigation bar with 'Overview' and 'Design' (highlighted) buttons, and a 'Step 2' button with a document icon.

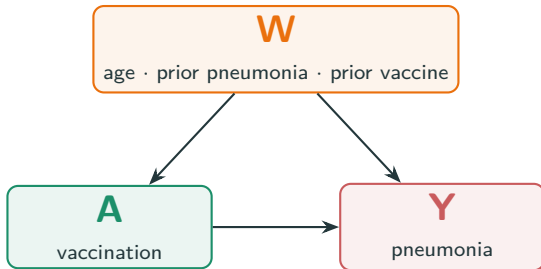
THE WORKSHOP VIEW

A compact model makes the confounding structure visible.

- Stipulated baseline set: age, prior pneumonia, and prior vaccination.
- Do not adjust for post-baseline descendants.
- The kit contains the full five-node, nine-edge DAG.

Accept, correct, or reject?

Does this adjustment set follow from the DAG we are willing to defend?



Collapsed discussion view; use the complete DAG for the audit.

Step 3: make the protocol computable

Observed data

3 Map the data

5,000 people

1,627

vaccinated

334 events

The fixed CSV has one row per person, complete baseline covariates, treatment, and 12-month outcome.

Protocol element

Population

Treatment

Outcome

Baseline W

Time zero

Observed-data mapping

One eligibility-defined row per participant

Binary baseline vaccination indicator A

Binary 12-month pneumonia hospitalization Y

Age, prior pneumonia, prior vaccination

Eligibility, treatment assignment, and follow-up aligned

Mapping check: Can every protocol element be operationalized without changing the question?

Can the observed data answer the causal question?

- exchangeability given W
- practical positivity
- consistency
- no interference

Three-part audit

Choose the weakest assumption. Name the evidence you would seek and one design change if it fails.

Assumption	Data can inform	Still requires judgment
Exchangeability	Covariate balance; proxies	Whether key common causes are measured
Positivity	Fitted score range; support	Whether treatment is possible in every target stratum
Consistency	Recorded treatment versions	Whether the intervention is well defined
No interference	Cluster/context clues	Whether one person's treatment affects another

For this exercise the generator makes the assumptions explicit; the observed data alone still cannot certify them.

CAUSAL TARGET

$$\psi = E[Y^1] - E[Y^0]$$

OBSERVED-DATA MAPPING

$$E_W[E(Y | A = 1, W) - E(Y | A = 0, W)]$$

Audit question

What changes if the population, outcome window, or censoring intervention changes?

The screenshot displays the 'Causal Navigator' web application. On the left, a 'Steps' sidebar lists the workflow: Overview, Define Causal Question, Specify Causal Model, Consider Observed Data, Assess Identifiability, Define Statistical Estimand (highlighted in orange), Choose Estimator, Sensitivity Analysis, and Results & Interpretation. The main panel shows the 'Statistical Estimand' section with the formula $\text{psi_RD} = E_W(E(Y | A=1, W) - E(Y | A=0, W))$ and a description of the risk ratio. Below this, the 'Link to Causal Under Identification' section explains the conditions for the statistical estimand to equal the causal effect. The 'Censoring Adjustment' section states that no adjustment is needed for the teaching dataset but is required for real studies. At the bottom, a progress bar shows 'Step 4' completed and 'Step 6' next, with a 'Design' button highlighted in the center.

Home Map

Causal Navigator

Export study summary AI Reset

Statistical Estimand

$\text{psi_RD} = E_W(E(Y | A=1, W) - E(Y | A=0, W))$, where $W = (\text{age}, \text{priorPneumonia}, \text{priorVaccine})$. The risk ratio divides the two standardized risks.

Link to Causal Under Identification

The statistical estimand equals the causal effect if conditional exchangeability given W , positivity, consistency/well-defined interventions, and no interference hold, and the target population and time zero are correctly represented.

Censoring Adjustment

The teaching dataset has complete 1-year outcomes, so no censoring adjustment is needed. In a real study, define 365-day risk under elimination of censoring and estimate inverse-probability-of-censoring weights or model the censoring mechanism.

< Step 4

0/4 Design

Step 6

6 Choose the estimator

- **Primary:** TMLE for the risk difference
- $\bar{Q}(A, W)$: outcome regression
- $g(W)$: treatment mechanism
- influence-function 95% CI

Boundary question

What can an estimator repair—and what can it never repair?

Outcome-blind work actually completed

No repeated simulation, bias, or RMSE study is included. The adjusted outcome result remains hidden until Steps 1–7 are locked.

Overlap diagnostic

Fitted parametric propensity scores range from 0.0765 to 0.9091. This is a diagnostic—not proof of structural positivity.

Reproducibility convention

Report the version-sensitive TMLE result rounded to -3.0 points (95% CI $-4.6, -1.5$).

Prespecify the stress tests before the result.

- unmeasured confounding
- outcome misclassification
- selection and censoring
- alternative learners and truncation

Choose one gap

What analysis or new evidence would make that gap more transparent?

Sampling uncertainty

The influence-function interval quantifies uncertainty for the statistical estimand under the fitted analysis.

Causal uncertainty

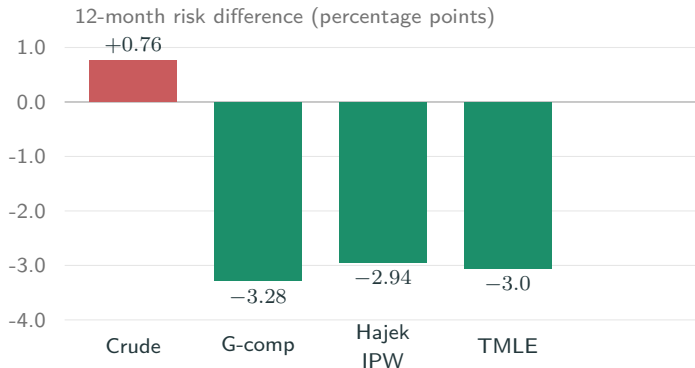
Separate analyses must address unmeasured confounding, measurement, selection, treatment versions, and interference.

Descriptive tipping point

A +3.0-point shift would move the TMLE point estimate to the null. This is not a formal E-value.

Adjustment reverses the sign

Result reveal



Synthetic teaching result — not clinical evidence.

Crude
+0.76 pp

TMLE
-3.0 pp

95% CI: -4.6 to -1.5 pp

What this shows

Adjusted analyses tell a different story under the stipulated model. Agreement does not prove the assumptions.

DEFENSIBLE

“In this synthetic exercise, adjustment reverses the crude association under the stipulated causal model.”

Supported claim

A statement about this workflow, this fixed dataset, and the assumptions explicitly recorded in the roadmap.

NOT DEFENSIBLE

“The vaccine prevents pneumonia in real patients.”

Still unknown

Clinical effectiveness, safety, transportability, real EHR bias, and population transmission effects.

Accept, correct, or reject?

Cite one Step 4 assumption and one limitation when revising the conclusion.

YOUR DECISION LOG

Choose one field from Steps 1–8.

1. What evidence makes the entry defensible?
2. What would make you revise it?
3. Which downstream decision changes if you revise it?

Pair protocol

Accept, correct, or reject the entry. Compare reasoning before export.

Participant kit

Participant workbook PDF
Complete DAGitty specification
Fixed synthetic dataset
Reproducible analysis script

Answer key—after the activity

Completed eight-step roadmap in Mark-down

Module F leaves us with:

- a specified target trial
- an explicit causal model
- an identification argument
- a statistical estimand
- a prespecified estimator and sensitivity plan

MODULE G HANDOFF

Use the same cohort and estimand to compare adjusted estimates, uncertainty, and the limits that remain.

Design first

Estimate second. Interpret last.